

Project Design Phase

Solution Architecture

Date	13 July 2026
Team ID	SWUID2026196344
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture for the Nutrition Assistant project defines the overall structure of the application and illustrates how different software components interact to deliver personalized nutrition management services. It serves as a blueprint that guides the development process, ensuring scalability, maintainability, security, and efficient communication between the frontend, backend, and database.

The primary objectives of the solution architecture are to:

- Design a secure and scalable MERN Stack application for nutrition management.
- Define the interaction between the React frontend, Express.js backend, MongoDB database, and REST APIs.
- Organize the application into well-structured modules such as User Authentication, Dashboard, Meal Planning, Nutrition Suggestions, and Profile Management.
- Ensure secure user authentication using JWT and protect sensitive user information.
- Provide a responsive, user-friendly, and maintainable system that supports personalized meal planning and nutrition tracking.

Example - Solution Architecture Diagram:

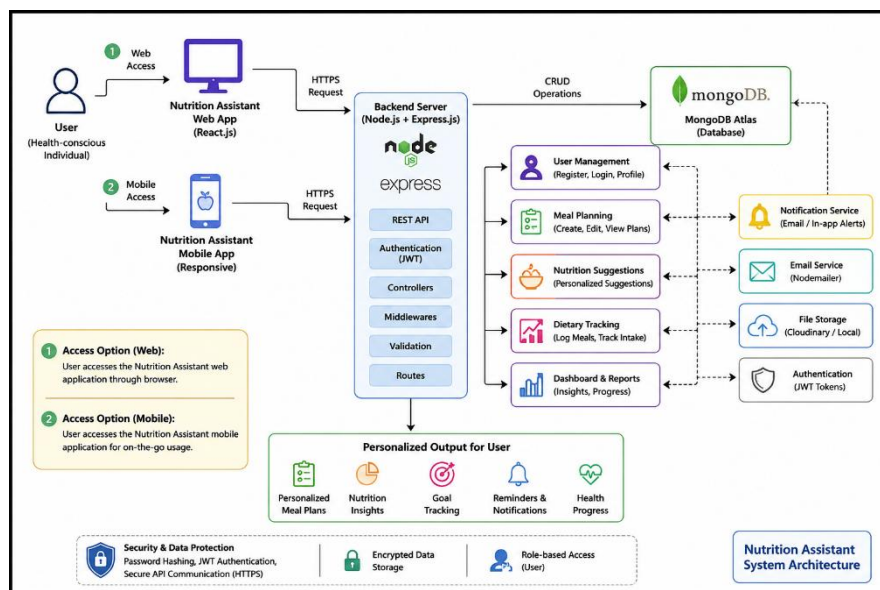


Figure 1: Architecture and data flow of the Nutrition Assistant Application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>